

Received 6 July 2024, accepted 31 July 2024, date of publication 5 August 2024, date of current version 14 August 2024.

Digital Object Identifier 10.1109/ACCESS.2024.3438992

RESEARCH ARTICLE

EnPredRNN: An Enhanced PredRNN Network for Extending Spatio-Temporal Prediction Period

DALI WU^{1,2}, JIAYANG KONG^{1,3}, ZHICHENG LI^{1,2}, GUOJING ZHANG^{1,2},
HUAICONG ZHANG^{1,2}, JING LIANG^{1,2}, AND XING ZHANG^{1,2}

¹School of Computer and Information Science, Qinghai Institute of Technology, Xining 810016, China

²College of Computer Technology and Application, Qinghai University, Xining 810016, China

³School of Marxism, Beijing Normal University, Beijing 100875, China

Corresponding author: Xing Zhang (2023740048@qhu.edu.cn)

This work was supported in part by the Natural Science Foundation of Qinghai Province under Grant 2023-QLGKLYCZX-017, Grant 2023-ZJ-906M, Grant 2023-QLGKLYCZX-026, and Grant 2023-QLGKLYCZX-027; and in part by Qinghai University 2023 Young Researchers' Fund under Grant 2023-QGY-7.

ABSTRACT We propose the Enhanced Predictive Recurrent Neural Network (EnPredRNN) based on PredRNN to extend the period of spatio-temporal prediction. To better capture global spatial dependencies, we integrate the Self-Attention (SA) module into PredRNN's basic unit, forming the Enhanced Long Short-Term Memory (EnLSTM). Aiming at the problem that the standard temporal memory state in PredRNN contains insufficient information about inter-frame motion, we propose the Enhanced Temporal Memory (ETM) by aggregating past multi-step temporal memory states. Aiming at the gradient vanishing problem in Recurrent Neural Networks (RNN), the Alleviating Gradient Vanishing (AGV) structure is used to construct the high-speed path that facilitates gradient propagation. Experimental results show that EnPredRNN effectively extends spatio-temporal prediction from ten to thirty time-steps.

INDEX TERMS Gradient vanishing, long-term prediction, spatio-temporal modeling.

I. INTRODUCTION

Due to the ability to process both temporal and spatial information, methods of spatio-temporal prediction are used in numerous fields. Traffic prediction [1], precipitation prediction [2] and video prediction [1] belong to the category of spatio-temporal prediction. Spatio-temporal prediction tasks are usually characterized by considerable complexity and randomness, which prevents traditional mathematical and theoretical methods from achieving good results in dealing with the relevant problems. Considering the good performance of deep learning models in modeling complex patterns, more and more studies adopt these methods for spatio-temporal prediction. In recent years, various spatio-temporal prediction models based on deep learning have emerged. Among them, deep learning models utilizing Recurrent Neural Networks (RNN) [4] as the main architecture have become the mainstream spatio-temporal prediction methods. RNN is the deep

learning model with recurrent structure, which is well suited for modeling temporal features in spatio-temporal problems. Therefore, it has become the basis for developing various spatio-temporal prediction models. Srivastava et al. [5] used Fully Connected Long Short-Term Memory (FC-LSTM) to learn the representation of video sequences and thus extrapolated the future. However, the fully connected layers in FC-LSTM cannot handle spatial information well. Shi et al. [6] proposed Convolutional LSTM (ConvLSTM), which extends FC-LSTM with the convolutional structure in both input-to-state and state-to-state transitions. Short-term precipitation forecasting experiments show that ConvLSTM is able to capture the spatio-temporal correlation better than FC-LSTM. Since ConvLSTM is only the simple fusion of spatial and temporal feature processing structures, it performs poorly in processing complex spatio-temporal information. The basic structural unit of Predictive Recurrent Neural Network (PredRNN) [7], Spatio-Temporal LSTM (ST-LSTM), can be viewed as consisting of two basic structural units of the ConvLSTM, which deal with temporal and spatial

The associate editor coordinating the review of this manuscript and approving it for publication was Yudong Zhang¹.

information, respectively. The ingenious design of PredRNN enables it to effectively model shape deformation and motion trajectories in spatio-temporal sequences. It is undeniable that these methods have achieved impressive results in the field of spatio-temporal prediction. However, most methods can only effectively predict spatio-temporal information for the next ten time-steps. When the prediction time expands, the prediction accuracy of existing models decreases drastically, which poses challenges in handling the task of long-sequence spatio-temporal prediction.

In recent years, with the development of neural networks [8], [9], the Transformer family [10], [11], [12] has achieved impressive results in the fields of image recognition [13], object detection [14], text generation [15], machine translation [16], and so on. As the core component of the Transformer models, the attention [17] mechanism has become the focus that researchers concern. Compared to the convolutional operation, the attention mechanism captures the global features of the image better, which is helpful in dealing with spatial information problems.

In order to extend spatio-temporal prediction period, we make some improvements on the basis of PredRNN and propose the EnPredRNN. In order to extract the spatial information more comprehensively and fully, we apply the Self-Attention (SA) module to the ST-LSTM, which is the basic constituent unit of PredRNN, and obtain the Enhanced LSTM (EnLSTM), which is the basic constituent unit of Enhanced PredRNN (EnPredRNN). Aiming at the problem that the standard temporal memory state in PredRNN contains insufficient and unreliable information about inter-frame motion, we effectively aggregate the past multi-step temporal memory states and propose the Enhanced Temporal Memory (ETM) state, which incorporates more information conducive to improving the quality of prediction images. Aiming at the gradient vanishing problem in RNN, the Alleviating Gradient Vanishing (AGV) structure is used to construct the high-speed path that facilitates the propagation of gradients. The experimental results demonstrate that the EnPredRNN can extend the spatio-temporal prediction to thirty time-steps into the future. The major contributions of the paper are as follows.

- We propose a variant of PredRNN, called EnPredRNN, for spatio-temporal prediction, which can effectively extend the prediction period.
- We design the Enhanced Long Short-Term Memory unit EnLSTM to capture global dependencies in spatial information better, propose the Enhanced Temporal Memory state ETM that aggregates past multi-step temporal memory states to improve the quality of predicted images, and employ the Alleviating Gradient Vanishing structure AGV to facilitate the propagation of gradients.
- We conducted short-time prediction experiments based on the MovingMNIST dataset and long-time prediction experiments based on MovingMNIST, KTH and Radar Echo datasets. Results show that the EnPredRNN not only achieves the best prediction performance, but

also effectively extends the duration of spatio-temporal prediction.

II. RELATED WORK

Since spatio-temporal prediction is one of the sequence modeling problems, RNN and its various variants have been used to cope with such tasks. Navares and Aznarte [18] used FC-LSTM to predict air quality in the Madrid area. Due to the ability to process temporal and spatial information simultaneously, models incorporating convolutional and recurrent structures have become important methods for spatio-temporal prediction. Shi et al [6] incorporated convolutional operations within LSTM [19] and proposed ConvLSTM, which achieved better results than FC-LSTM in short-term precipitation prediction experiments. Zapata-Impata et al. [20] employed ConvLSTM to learn spatio-temporal tactile features for slip direction detection. Çiçek and Gören [21] predicted the remaining battery capacity of the smartphone for the next 24 hours using ConvLSTM. In order to obtain clear video prediction results, Su et al. [22] adopted a new tensor training module on ConvLSTM and proposed a fully convolutional higher-order LSTM model, Conv-TT-LSTM. GRU [23] is a variant of LSTM. Due to its relatively simple structure, GRU is computationally more efficient. ConvGRU [24] is the model that incorporates the convolution operation inside the GRU, and its structure is similar to ConvLSTM. Zhang et al. [25] used ConvGRU for the prediction of vegetation dynamics on the plateau. Zhang et al. [26] used ConvGRU to predict the velocity of mountain glacier movement along the Qinghai-Tibet Railway. Considering the intrinsic interactions between current inputs and previous outputs, He et al. [27] improved the ConvGRU and proposed M-ConvGRU. They used the model for short-term precipitation prediction based on radar echoes. TrajGRU [28] enhances the ability to model and predict the trajectories of moving objects by introducing the trajectory tracking structure. Zeng et al. [29] used TrajGRU for the prediction of spatio-temporal sequences of radar echoes. Zhang et al. [30] used TrajGRU to improve the accuracy of wind speed forecasting. ConvLSTM, Conv-TT-LSTM, ConvGRU, M-ConvGRU, and TrajGRU fail to efficiently fuse convolutional and cyclic structures, rendering them incapable of efficiently processing complex spatio-temporal information. PredRNN [7] adopts temporal memory state and spatio-temporal memory state to capture the temporal dynamic features and spatial relationships in spatio-temporal sequences, respectively, so as to achieve the prediction results closer to the truth. Zang et al [31] used PredRNN for atmospheric visibility prediction. Feng et al. [32] used PredRNN for the prediction of the Arctic sea ice concentration. PredRNN-V2 [33] further improves on PredRNN, which employs the memory decoupling loss, which prevents temporal and spatio-temporal memory states from learning redundant features.

Zhang and Wu [34] performed video prediction using PredRNN-V2. There is no doubt that the above models

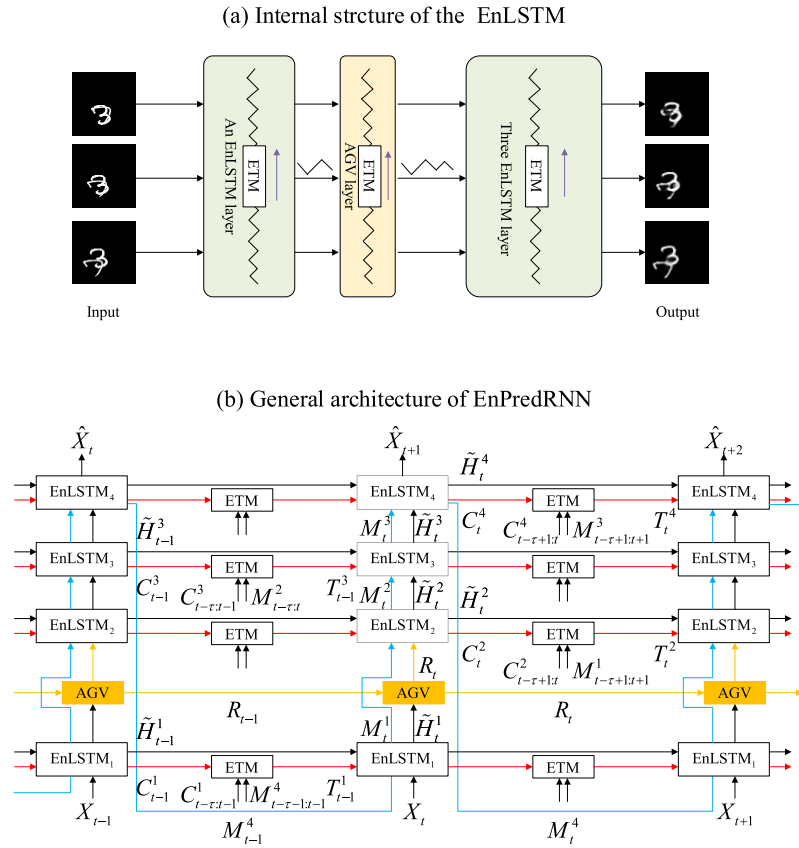


FIGURE 1. (a) Generalized representation of the EnPredRNN network architecture. This figure provides a simple and clear reflection of the process by which the input images are processed by the EnLSTM layer, the AGV layer, and the ETM structure to obtain the output images. (b) Detailed representation of the EnPredRNN network architecture. The figure shows the specific details of the network and the process of spatio-temporal information flow. The difference with PredRNN is that the architecture of EnPredRNN contains the addition of the ETM module and the AGV structure. In particular, the AGV structure exists only between the first and second networks.

have achieved impressive results in spatio-temporal prediction. However, many researchers have only predicted spatio-temporal information for the next ten time-steps. In the case of increasing time span, the difference between the results predicted by the existing models and the truth is obvious. The wide application of attention mechanism significantly improves the model's processing of images. Many researchers have developed various deep learning models incorporating the attention mechanism in the areas of image recognition, object detection, and image generation. These models beat the original classical models on numerous tasks. Zhou et al. [35] achieved higher accuracy than ShuffleNet V2 [36], MobileNet V2 [37] and AlexNet [38] when using the improved ShuffleNet V2 incorporating the channel attention mechanism for rice leaf disease recognition. The improved RetinaNet embedded with the dual-pooled attention module demonstrated better performance than the original model when performing pedestrian detection [39]. Tang et al. [40] proposed a novel AttentionGAN, which is capable of generating more realistic images than CycleGAN [41], DualGAN [42] and DiscoGAN [43]. The attention

mechanism enables the model to be more flexible and perceptive by giving different attention to different regions, thus achieving excellent results in various image processing tasks.

In order to perform longer spatio-temporal prediction while maintaining accuracy, we design the EnPredRNN. Inspired by the fact that the attention mechanism can improve the model's ability to process image information, EnLSTM, the basic constituent unit of EnPredRNN, is formed by incorporating the SA module into ST-LSTM, the basic constituent unit of PredRNN. The SA module facilitates the extraction of more comprehensive information. Compared to the standard temporal memory state in PredRNN, the ETM state we propose aggregates the temporal memory state of the past multi-steps and contains more adequate inter-frame motion information. In addition, we construct the high-speed path that facilitates gradient propagation to cope with the gradient vanishing problem in RNN. Experiments show that EnPredRNN achieves better results than other models while predicting spatio-temporal information for the next thirty steps.

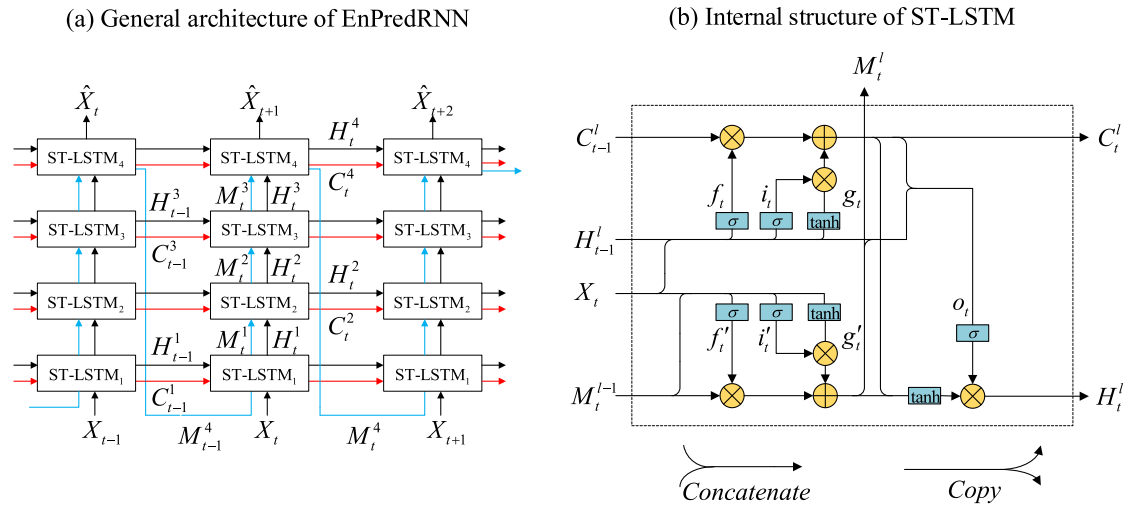


FIGURE 2. (a) General architecture of PredRNN. The model consists of four layers of the networks. The standard temporal memory state C is propagated horizontally within the layers, and the spatio-temporal memory state M is propagated in a zigzag pattern between the layers. These two states enable the PredRNN to process temporal and spatial information separately. (b) Internal structure of ST-LSTM. It can be clearly seen that the ST-LSTM contains six gating mechanisms $i_t, g_t, f_t, i'_t, g'_t$ and f'_t , and their division of labor and cooperation endow the PredRNN with powerful spatio-temporal information processing capabilities.

The rest of the paper is organized as follows. Section III gives the foundation on which the EnPredRNN is grounded, i.e., the introduction of the PredRNN, and the principles of the three important components of the model, namely, EnLSTM, ETM, and AGV. In Section IV, we test the performance of the EnPredRNN against other existing models using both long-term and short-term prediction experiments. Finally, the corresponding conclusions are drawn and some discussions are presented in Section V.

III. METHODS

A. GENERAL ARCHITECTURE

Since the EnPredRNN is based on the PredRNN, the network architecture of EnPredRNN is similar to PredRNN. Fig. 1 illustrates the general architecture of the model. For easy understanding, the general process of spatio-temporal information flow in the network is graphically shown in Fig. 1 (a). Predicted results can be obtained by passing the input information through the first EnLSTM layer, the AGV layer and the last three EnLSTM layers respectively. The spatio-temporal information is represented by waveform line segments, which flow horizontally between the layers as shown by the black arrows, and vertically within the layers as shown by the purple arrows. The ETM aggregates the spatio-temporal information within the layers into the Enhanced Temporal Memory state. Fig. 1 (b) is the more specific representation of Fig. 1 (a). The figure depicts the more detailed architecture of the EnPredRNN, showing the flow of spatiotemporal information. It can be seen that the basic constituent structure of the EnPredRNN is the EnLSTM, which is obtained by applying the SA module to the hidden state H_t in ST-LSTM. An ETM module is embedded between

every two neighboring EnLSTM units, which transforms the standard temporal memory state output from the previous time step into the enhanced temporal memory state as the input for the current time step. In order to adequately mitigate the gradient vanishing, the AGV is placed between the first and second layers of the network.

B. BASE MODEL

PredRNN is the deep learning model for spatio-temporal prediction proposed by Wang et al. [7]. The overall architecture of PredRNN is shown in Fig. 2 (a). It can be seen that the predicted results can be obtained from the input images processed by the four ST-LSTM network layers. The two memory states inside the PredRNN are the standard temporal memory state C and the spatio-temporal memory state M . The standard temporal memory state is propagated between different memory units in the same layer, realizing the process of transferring temporal information from the historical moment to the current moment. The temporal memory state is propagated throughout the network in the zigzag shape. It first passes vertically from the first layer of the current time step to the last layer, and then from the last layer to the first layer of the next time step. The passing of the spatio-temporal memory state strengthens the network's utilization of spatio-temporal information. The basic structural unit of PredRNN, ST-LSTM, can be viewed as the fusion of two basic structural units of ConvLSTM. The internal structure of ST-LSTM is shown in Fig. 2 (b). ST-LSTM contains six gating mechanisms: input gate, input modulation gate, output gate, new input gate, new input modulation gate and new output gate. Under their joint action, new temporal state C_t and spatio-temporal state M_t are obtained by the

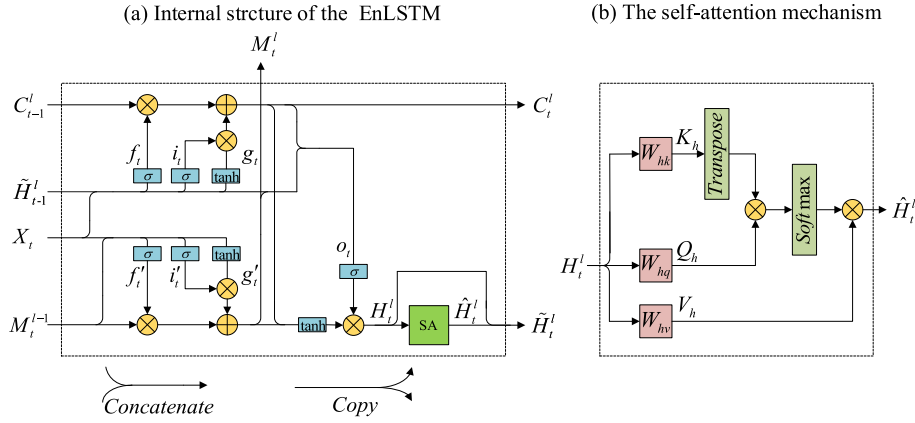


FIGURE 3. (a) Internal structure of EnLSTM. EnLSTM is formed by incorporating the SA mechanism in ST-LSTM. EnLSTM owns stronger information processing ability than ST-LSTM. (b) The operating principle of the SA mechanism. Through a series of operations of Q_h , K_h and V_h , the SA mechanism enables the model to fully extract the features of the image.

transformation of the temporal state C_{t-1} , spatio-temporal state M_{t-1} and other corresponding states of the last moment. The formula of PredRNN is as follows.

$$\begin{aligned}
 g_t &= \tanh(W_{xg} * X_t + W_{hg} * H_{t-1}^l + b_g) \\
 i_t &= \sigma(W_{xi} * X_t + W_{hi} * H_{t-1}^l + b_i) \\
 f_t &= \sigma(W_{xf} * X_t + W_{hf} * H_{t-1}^l + b_f) \\
 C_t^l &= f_t \circ C_{t-1}^l + i_t \circ g_t \\
 g'_t &= \tanh(W'_{xg} * X_t + W'_{mg} * M_{t-1}^{l-1} + b'_g) \\
 i'_t &= \sigma(W'_{xi} * X_t + W'_{mi} * M_{t-1}^{l-1} + b'_i) \\
 f'_t &= \sigma(W'_{xf} * X_t + W'_{mf} * M_{t-1}^{l-1} + b'_f) \\
 M_t^l &= f'_t \circ M_{t-1}^{l-1} + i'_t \circ g'_t \\
 o_t &= \sigma(W_{xo} * X_t + W_{ho} * H_{t-1}^l + W_{co} * C_t^l \\
 &\quad + W_{mo} * M_t^l + b_o) \\
 H_t^l &= o_t \circ \tanh(W_{1 \times 1} * [C_t^l, M_t^l])
 \end{aligned} \quad (1)$$

where X_t denotes the input of the model at time t . C_t and M_t denote the standard temporal memory state and the spatio-temporal memory state at time t , respectively. H_t denotes the hidden state at moment t . b denotes the bias value. i_t , g_t and f_t denote input gates, input modulation gates and output gates respectively. i'_t , g'_t , f'_t denote new input gates, input modulation gates and output gates, respectively. o_t denotes the output gate. l denotes which layer of the network the corresponding state is located. $*$ denotes the convolution, and $\circ$ denotes the Hadamard product. σ denotes the sigmoid function.

C. ENHANCED LSTM

Enhanced LSTM is the basic constituent structure of EnPredRNN, which is obtained by applying the self-attention

mechanism to the hidden state H_t in ST-LSTM. The internal structure of EnLSTM is shown in Fig. 3 (a). The SA mechanism allows the model to dynamically focus on important information of other locations while dealing with individual locations of the hidden state, which helps the model to comprehend the overall image instead of focusing only on local features. The operation of the SA mechanism is shown in Fig. 3 (b). The formula for extracting the global information of the hidden state using the SA mechanism is as follows.

$$\begin{aligned}
 Q_h &= W_{hq} * H_t^l \\
 K_h &= W_{hk} * H_t^l \\
 V_h &= W_{hv} * H_t^l \\
 Sco &= Q_h \cdot K_h^T \\
 Wei &= \text{Soft max}(Sco) \\
 \hat{H}_t^l &= Wei \cdot V_h
 \end{aligned} \quad (2)$$

where Q_h , K_h and V_h denote the query matrix, the key matrix and the value matrix, respectively, which all originate from the hidden state H_t . The attention score Sco is obtained by computing the similarity between Q_h and K_h . $\cdot$ denotes matrix multiplication. The attention weight Wei can be obtained by performing the *Soft max* operation on the similarity score. The new hidden state $\hat{H}_t$ can be regarded as the global key information extracted from the value matrix using the attention weight matrix. The final hidden state $\hat{H}_t$ is denoted as follows.

$$\tilde{H}_t^l = \tilde{W} * [H_t^l, \hat{H}_t^l] \quad (3)$$

It is the aggregation of the original hidden state H_t and the new hidden state $\hat{H}_t$.

D. ENHANCED TEMPORAL MEMORY MODULE

The skeleton of EnPredRNN is the recurrent neural network. The information memory function of the recurrent neural network lies in the fact that the output of each node is not

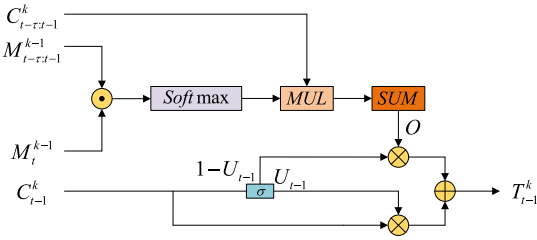


FIGURE 4. Internal structure of the ETM module. The ETM module is capable of aggregating a series of standard temporal memory states from the historical time steps into the ETM state T_{t-1} , as the input of the current time step. The module extends the temporal sensing field of the nodes of the model, allowing the model to better capture the temporal dependency in the sequence.

only related to the output of the nodes in the previous layer at the same time step, but it is also affected by the output of the nodes at the previous time step in the same layer. However, the output of the previous time step contains only limited time-dependent information, which can affect the model's ability to express spatio-temporal patterns. To solve the problem, more useful historical information needs to be used as input of the current time step. The idea can be realized in two steps. The first step is to determine the search scope of the historical information, and the second step is to design the scheme for extracting useful information from the historical information. In response to the first step, since historical information contains outputs from every past time step, one possible approach is to use all outputs as the search scope. Such approach ensures the comprehensiveness of the information, but poses numerous problems. First, over large search range will cause enormous increase in computational demand, which in turn affects the practicality of the model; second, the output of the current time step is more related to the nearby historical information and less related to the distant historical information, and the large range of information search not only leads to less significant improvement in the model performance, but also introduces much noise. Therefore, the search range of historical information should be set to the appropriate value. To achieve the second step, the importance of the outputs of different time steps is calculated according to the impact of the selected range of historical information on the current outputs, and then they are organically aggregated into the input of the current time step according to their importance. In this paper, we propose to use information of ten historical time steps to predict information of thirty future time steps. The search range for historical information is set to the outputs of five time steps near the current moment. As the basis of the proposed model, the two important memory states in PredRNN are the standard temporal memory state C_t and the spatio-temporal memory state M_t . The standard temporal memory state C_t propagates along the direction of loop unfolding and it stores the time-dependent information in the sequence. In order for the model to better capture the temporal dependency in the sequence, the Enhanced Temporal Memory module is designed in the paper. The internal structure

of the module is shown in Fig. 4, which can aggregate a series of standard time memory states from the historical time steps into the Enhanced Temporal Memory state T_{t-1} , which is used as the input of the current time step. Specifically, considering that the spatio-temporal memory state M_t contains rich spatio-temporal information in the output of the node, the paper employs the correlation between the corresponding spatio-temporal states to measure the degree of importance of different standard temporal memory states. According to the Importance, the standard temporal memory states are rationally combined together to form the fusion state O . Since the standard time memory state of the previous time step is closely related to the output of the current time step, the standard time memory state C_{t-1} of the previous time step is used to strengthen the fusion state O to obtain the strengthened time memory state T_{t-1} . The formula for the ETM state is as follows.

$$\begin{aligned}
 q_{t-i} &= SUM \left(M_{t-i}^{k-1} \circ (W_{qm} * M_t^{k-1}) \right) \\
 \alpha_{t-j} &= \frac{q_{t-j}}{\sum_{i=1}^{\tau} q_{t-i}} \\
 O &= \sum_{j=1}^{\tau} \alpha_{t-j} \cdot C_{t-j}^k \\
 U_{t-1} &= \sigma \left(W_{uc} * C_{t-1}^k \right) \\
 T_{t-1}^k &= U_{t-1} \circ C_{t-1}^k + (1 - U_{t-1}) \circ O \quad (4)
 \end{aligned}$$

where q_{t-i} denotes the product of the spatio-temporal memory state at moment $t - i$ and the spatio-temporal memory state at moment t , which is used to compute the significance α_{t-i} of the standard temporal memory state at moment $t - i$. U_{t-1} is the confluence gate, which converges the standard temporal memory state C_{t-1}^k and the fusion state O together into the ETM state T_{t-1}^k .

E. ALLEVIATING GRADIENT VANISHING STRUCTURE

Recurrent neural networks achieve parameter updating during training by the time-based backpropagation algorithm. Similar to the principle of backpropagation algorithm used in normal feedforward neural networks, the essence of the time-based backpropagation algorithm is the gradient descent method. It has revealed that ordinary feedforward neural networks suffer from gradient vanishing as the layers of the net deepen. This is related to the concatenation effect in the backpropagation algorithm. The same problem exists with recurrent neural networks due to the similar gradient updating strategy. Gradient vanishing can make it difficult to update the parameters and affect the learning performance of the model. In particular, when the sequence to be learned is long, the recurrent neural network tends to forget the past information.

Plagued by the problem of vanishing gradients, most spatio-temporal prediction models shorten the prediction duration. However, the approach does not solve the problem essentially and limits the practical scope of the models.

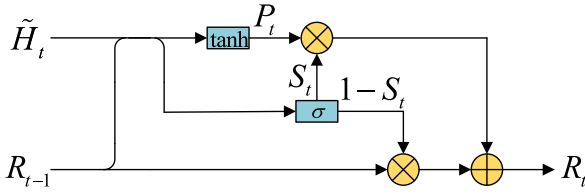


FIGURE 5. Internal details of the AGV structure. The AGV structure provides a high-speed path for the propagation of gradients. In order to balance the model performance and computational resources, the AGV structure is only embedded between the first and second layers of the network in the paper.

In order to cope with the gradient vanishing problem and extend spatio-temporal prediction period, the Alleviating Gradient Vanishing structure is designed in the paper, which provides a convenient channel for the propagation of the gradient. Fig. 5 shows the internal details of the AGV structure. The specific formula of the structure is as follows.

$$\begin{aligned} P_t &= \tanh(W_{ph} * \tilde{H}_t + W_{pr} * R_{t-1}) \\ S_t &= \sigma(W_{sh} * \tilde{H}_t + W_{sr} * R_{t-1}) \\ R_t &= S_t \circ P_t + (1 - S_t) \circ R_{t-1} \end{aligned} \quad (5)$$

where P_t is the fused hidden state at moment t . S_t is the transition gate, which converts the high-speed memory state R_{t-1} at moment $t - 1$ to the high-speed memory state R_t at the current moment. The high-speed memory state propagates through the gradient convenience path, which promotes the model's learning of long-distance-dependent information and contributes to alleviating gradient vanishing problem.

In order to better model spatio-temporal patterns, the deep network model is constructed in the paper. Theoretically, as the module specially designed for cyclic structures, the AGV structure can be used for every layer of the network in the model. However, this approach poses some problems. Excessive use of the AGV structure can only improve the model performance limitedly, but it will waste a lot of computational resources. Therefore, rationally combining the AGV structure with the model is an optimal choice. Since the original inputs have greater influence on the prediction results than the features extracted from the deeper network, the structure is embedded between the first and second layers of the network. The whole model after embedding can be expressed as follows.

$$\begin{aligned} &\tilde{H}_t^1, C_t^1, M_t^1 \\ &= \text{EnLSTM}_1(X_t, \tilde{H}_{t-1}^1, C_{t-1}^1, M_{t-1}^L, C_{t-\tau:t-1}^1, M_{t-\tau-1:t-2}^L) \\ &R_t \\ &= \text{AGV}(\tilde{H}_t^1, R_{t-1}) \\ &\tilde{H}_t^2, C_t^2, M_t^2 \\ &= \text{EnLSTM}_2(R_t, \tilde{H}_{t-1}^2, C_{t-1}^2, M_t^1, C_{t-\tau:t-1}^2, M_{t-\tau:t-1}^1) \end{aligned}$$

$$\begin{aligned} &\tilde{H}_t^k, C_t^k, M_t^k \\ &= \text{EnLSTM}_k(\tilde{H}_t^{k-1}, \tilde{H}_{t-1}^k, C_{t-1}^k, M_t^{k-1}, C_{t-\tau:t-1}^k, M_{t-\tau:t-1}^{k-1}) \end{aligned} \quad (6)$$

where L denotes the number of layers of the EnLSTM structural unit, and EnLSTM_k denotes the EnLSTM structural unit in layer k . AGV denotes the Alleviating Gradient Vanishing structure.

The yellow rectangle in Fig. 1 (b) shows where the AGV structure is embedded in the model, and the yellow line shows the convenient path from the first time step to the last time step. This path provides a high-speed route for the propagation of the gradient and facilitates rapid updating of the model parameters during training. The synergistic work of the AGV structure and the model provides a reliable scheme for the processing of long sequence information.

IV. EXPERIMENTS

A. DATASETS

In order to validate the spatio-temporal prediction ability of the model, we conducted experiments on MovingMNIST, KTH, and Radar Echo datasets. MovingMNIST is the dataset constructed from moving handwritten digits. KTH contains different kinds of continuous human movements. The Radar Echo dataset consists of radar echo images with fixed observation intervals. All three datasets contain complex spatio-temporal patterns that can be used to test the performance of the model.

1) MOVINGMNIST

MovingMNIST is the extension of MNIST, a dataset of handwritten digitized images. By applying the motion transform to the figures in MNIST, image sequences with motion properties can be generated. MovingMNIST consists of many such image sequences. Corresponding to different experimental scenarios in the paper, two MovingMNIST datasets are produced in the paper. The sequence in one dataset contains 20 frames of images. The sequence in the other dataset contains 40 frames of images. Both datasets contain 10,000 sequences in the training set and 5,000 sequences in the test set.

2) KTH

KTH contains six different human movement categories, namely walking, jogging, running, boxing, waving, and clapping. The images in the dataset are captured in different scenarios, including outdoor and indoor. We modeled the dataset for the experiment after MovingMNIST. The dataset contains a series of sequences of images of length 40. The first 10 images in the sequence are used to predict the next 30 images. The size of each image is 128×128 . The training and test sets contain 3389 and 2100 sequences respectively.

3) RADAR ECHO

The Radar Echo dataset is composed of numerous sequences of radar echo images. We collected radar echo data from

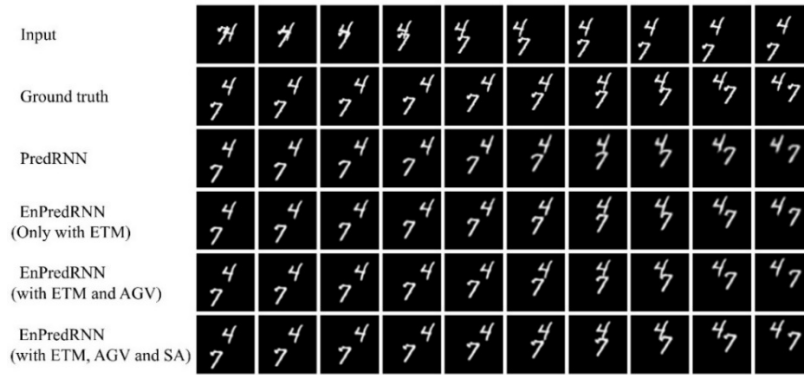


FIGURE 6. Qualitative comparison results of the four models. The first row shows the ten images of the input and the second row shows the ten real images in the future. The remaining rows show the prediction results of each model. The comparison results clearly reflect the positive enhancement of the prediction results by each improved part of the model.

Yinchuan area of Ningxia for two years. Through selection, noise reduction and cropping, we obtained a large number of time-continuous radar echo images. The image size is 200×200 . In order to obtain sufficient sequences, we segment the consecutive radar echo pictures using the sliding window. The length of the sliding window is set to 40, and the corresponding dataset consists of sequences, each containing 40 consecutive radar echo images. The first 10 frames in the sequence are used for learning and the last 30 frames are for prediction. We divided the training set and test set evenly in the ratio of 4:1. The training and test sets contain 5640 and 1410 sequences, respectively.

B. EVALUATION INDICATOR

While comparing the experimental results of each model on MovingMNIST and KTH datasets, Mean Square Error (MSE), Learned Perceptual Image Patch Similarity (LPIPS), Peak Signal to Noise Ratio (PSNR), and Structural Similarity Index Measure (SSIM) are used as the evaluation metrics in the paper. LPIPS, PSNR, SSIM evaluate the similarity of two images from different perspectives. MSE can be used to measure the difference between two images. When comparing the experimental results of each model on the Radar Echo dataset, the Critical Success Index (CSI), Heidke Skill Score (HSS), SSIM and MSE are chosen as the evaluation metrics in the paper. CSI and HSS are the comprehensive metrics that take into account both the Probability Of Detection (POD) and False Alarm Rate (FAR), which can evaluate the prediction results comprehensively. The formulas of CSI and HSS are as follows.

$$\text{CSI} = \frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$$

$$\text{HSS} = \frac{2(\text{TP} \times \text{TN} - \text{FN} \times \text{FP})}{(\text{TP} + \text{FN})(\text{FN} + \text{TN}) + (\text{TP} + \text{FP})(\text{FP} + \text{TN})} \quad (7)$$

where TP denotes (prediction = 1, truth = 1), FP denotes (prediction = 1, truth = 0), TN denotes (prediction = 0, truth = 0), FN denotes (prediction = 0, truth = 1). The larger the value of

TABLE 1. Quantitative comparison results for the four models.

Model	LPIPS↓	PSNR↑	SSIM↑	MSE↓
PredRNN [7]	0.09154	19.4	0.8729	55.29
EnPredRNN(Only with ETM)	0.08111	19.79	0.8834	50.89
EnPredRNN(with ETM and AGV)	0.0648	20.27	0.896	47.21
EnPredRNN(with ETM, AGV and SA)	0.05541	20.6	0.9019	44.44

CSI, the better the prediction of the model. The same is true for HSS.

C. IMPLEMENTATION DETAIL

Multiple sets of experiments are conducted in the paper to validate the predictive effectiveness of EnPredRNN. To ensure the fairness of the comparison results, we keep most of the parameter settings consistent across the models. All models employ 4-layer networks and the number of convolutional kernels per layer is 128. all input-to-state and state-to-state convolutional kernel sizes are 5×5 . the learning rate is set to 0.0001 and the batchsize is 8. We use Adam as the optimizer and perform 8000 iterations. Notably, some of the parameter settings in EnPredRNN are changed in order to balance the prediction quality and training efficiency. For EnPredRNN, the search range of historical information is set to 5 and the learning rate is 0.00003. All models are completed using PyTorch and run on NVIDIA GeForce GTX 1080 Ti GPUs. CUDA version is 11.5 and PyTorch version is 1.7.

D. ABLATION STUDY

In order to evaluate the contribution of each improvement component to enhance the predictive ability of the model, ablation experiments are conducted in the paper on the MovingMNIST dataset containing short sequences. The procedure of the experiment is to predict 10 consecutive digital motion pictures in the future using 10 consecutive digital

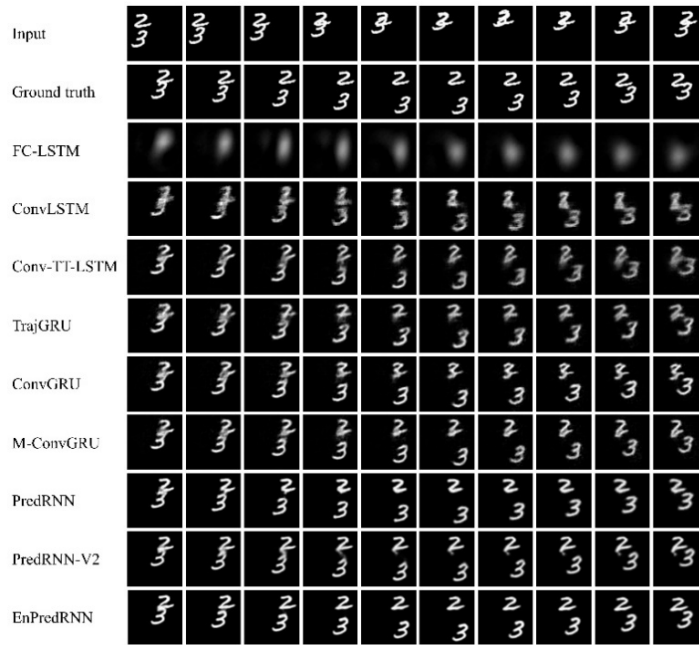


FIGURE 7. Presentation of Short-Term Prediction Results for nine models on MovingMNIST. The figure is a randomly selected sample from the prediction results of multiple models on the MovingMNIST dataset containing short sequences. The sample demonstrates the ability of each model to predict the movement of the numbers “2” and “3.”

motion pictures of input. The models involved in the experiment include the original PredRNN, the EnPredRNN only with ETM, the EnPredRNN with ETM and AGV, and the full version of the EnPredRNN that incorporates ETM, AGV, and SA. The design of the ablation study takes into account all the improvements and is highly convincing. Table 1 shows the quantitative results for the four models. Fig. 6 shows a representative comparison sample selected from the test set.

As can be seen from Table 1, among the four models, PredRNN has the worst prediction results, and all the evaluation metrics of the model are worse than the other models. The model that contains more improvements has better prediction results, and the full version of EnPredRNN has the best prediction results. It shows that all the improvements are effective and can improve the accuracy of the model’s prediction results. In general, the AGV structure provides the most significant improvement in model effectiveness. Compared with the EnPredRNN model only with ETM, the network with the addition of AGV shows substantial improvement in the performance of all the metrics. It reflects the important role of the AGV structure and justifies the subsequent use of the model for long sequence prediction tasks. As can be obtained from Fig. 6, all models can accurately predict the position of the digits due to the relatively simple movement pattern of the two digits in the figure. From frame 5 onwards, the prediction results of PredRNN are gradually blurred, while the numbers predicted by the other models remain clear. It shows that all the improvements are favorable for long distance prediction. In the prediction results, PredRNN

TABLE 2. Short-term prediction results of the nine models on the MovingMNIST dataset.

Model	LPIPS↓	PSNR↑	SSIM↑	MSE↓
FC-LSTM [5]	0.6081	15.52	0.4685	119
ConvLSTM [6]	0.1464	17.9	0.8035	72.7
Conv-TT-LSTM [22]	0.1323	18.21	0.8087	64.6
TrajGRU [28]	0.1268	18.7	0.7778	63.18
ConvGRU [24]	0.09937	18.82	0.8004	62.4
M-ConvGRU [27]	0.09658	19.01	0.8215	59.31
PredRNN [7]	0.09154	19.4	0.8729	55.29
PredRNN-V2 [33]	0.06924	20.24	0.8941	47.5
EnPredRNN	0.05541	20.6	0.9019	44.44

loses much of the edge information although it predicts the basic shapes of the two digits. Models that contain more improvements are more able to finely predict the details of the appearance of the digits in future frames. This is as expected.

E. SHORT-TERM PREDICTION EXPERIMENTS

In order to test the spatio-temporal prediction ability of EnPredRNN, a series of comparison experiments are conducted on the MovingMNIST consisting of short sequences. The length of the sequences in the dataset is 20, where the first 10 sheets are used as the input and the last 10 sheets are used as the prediction. In the experiments, EnPredRNN is compared with FC-LSTM, ConvLSTM, Conv-TT-LSTM, TrajGRU, ConvGRU, M-ConvGRU, PredRNN and PredRNN-V2.

Table 2 shows the short-term prediction results of the nine models on the MovingMNIST dataset. From the table,

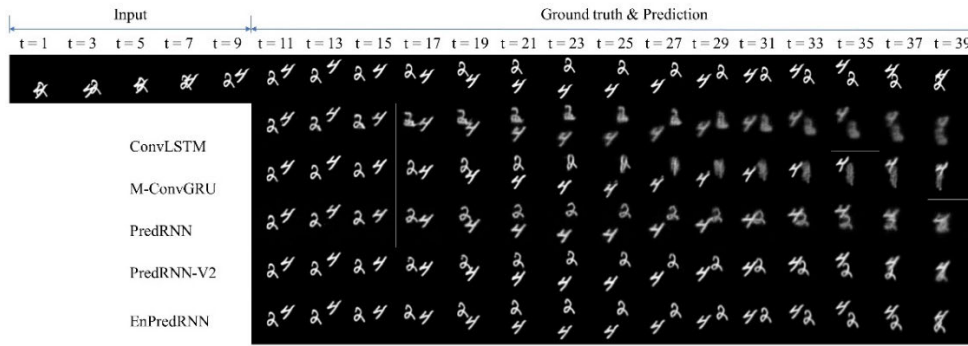


FIGURE 8. Results of qualitative experiments of the five models on MovingMNIST. Since the prediction sequence is too long, this figure shows the results of the five models predicting 30 images in the future using 10 images in the past at 1-frame intervals. The five models are selected based on the previous experimental results.

it can be seen that FC-LSTM has the worst prediction results and EnPredRNN has the best prediction results. The prediction results of PredRNN, PredRNN-V2 and EnPredRNN are significantly better than those of FC-LSTM, ConvLSTM, Conv-TT-LSTM, TrajGRU, ConvGRU and M-ConvGRU. EnPredRNN outperforms the other models in all the metrics. It shows that EnPredRNN has strong spatio-temporal prediction ability. Fig. 7 shows the short-term prediction results of seven models including EnPredRNN on MovingMNIST. It can be seen from the figure that FC-LSTM has the worst prediction results, and the other models can accurately predict the positions of the numbers “2” and “3.” For the details of the shapes of the two figures, the prediction results of ConvLSTM, Conv-TT-LSTM and TrajGRU are severely blurred, the results of ConvGRU and M-ConvGRU are obviously deformed, and the results of PredRNN, PredRNN-V2 and EnPredRNN are all close to the truth. Detailed observation reveals that EnPredRNN’s prediction of digital shapes is closest to the truth and the output of this model is the most accurate. The experimental results demonstrate the suitability of EnPredRNN to handle spatio-temporal modeling tasks.

F. LONG-TERM PREDICTION EXPERIMENTS

In order to verify the ability of EnPredRNN to predict spatio-temporal information for the long-term future, a series of comparative experiments are conducted in the paper on MovingMNIST, KTH and radar echo datasets containing long sequences. From the previous experiments, we see that the spatio-temporal prediction ability of PredRNN, PredRNN-V2 is much stronger than FC-LSTM, ConvLSTM, Conv-TT-LSTM, TrajGRU, ConvGRU and M-ConvGRU. Therefore, in the long-term prediction experiments, we adopt PredRNN and PredRNN-V2 as the main comparison models, and ConvLSTM and M-ConvGRU as the auxiliary comparison models representing the other models.

1) EXPERIMENTS ON MOVINGMNIST AND KTH

Each sequence in the MovingMNIST and KTH datasets contains 40 images, where the first 10 are used as inputs and

TABLE 3. Quantitative experimental results of the five models on the MovingMNIST dataset.

Model	LPIS↓	PSNR↑	SSIM↑	MSE↓
ConvLSTM [6]	0.3327	16.53	0.6132	106.31
M-ConvGRU [27]	0.2943	16.94	0.6536	91.42
PredRNN [7]	0.1754	18.01	0.7634	73.66
PredRNN-V2 [33]	0.1049	19.02	0.8579	64.21
EnPredRNN	0.09951	19.53	0.8654	58.08

TABLE 4. Performance comparison between EnPredRNN and the other two models on the KTH dataset.

Model	LPIS↓	PSNR↑	SSIM↑	MSE↓
PredRNN [7]	0.2037	26.99	0.7929	48.64
PredRNN-V2 [33]	0.2287	27	0.784	46.93
EnPredRNN	0.2015	27.15	0.8	46.73

the last 30 as predictions. The specifics of the two datasets have been described in detail earlier. Comparative experiments are conducted on these two datasets. the experimental models on the MovingMNIST dataset include ConvLSTM, M-ConvGRU, PredRNN, PredRNN-V2, and EnPredRNN, and the models on the KTH dataset are PredRNN, PredRNN-V2, and EnPredRNN, respectively.

Table 3 shows the quantitative experimental results of the five models on the MovingMNIST dataset. It can be seen from the table that compared to the other two models, the performance of PredRNN, PredRNN-V2, and EnPredRNN improves significantly on the four metrics. Among the three models, PredRNN has the worst performance, PredRNN-V2 outperforms PredRNN, and EnPredRNN has the best performance. It indicates that EnPredRNN has strongest ability to model the motion of long sequences. Fig. 8 shows the qualitative experimental results of all models on MovingMNIST. In order to get more intuitive visual effects, the results of the five models predicting 30 images in the future using 10 images in the past are shown in with 1 frame as an interval. As can be seen from the figure, the last half of the prediction results of ConvLSTM seriously lose the detail information, and the prediction of the number “2” by M-ConvGRU turns

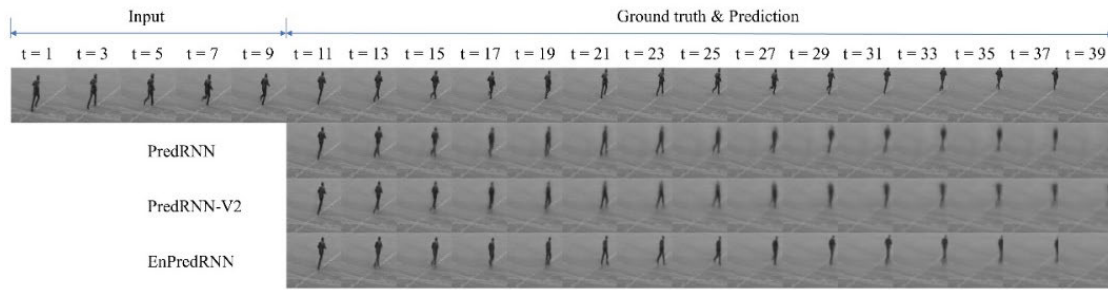


FIGURE 9. Comparison of representatively generated images for the three models. the predictions on the KTH dataset are presented in the same way as the results on MovingMNIST, which contains long sequences.

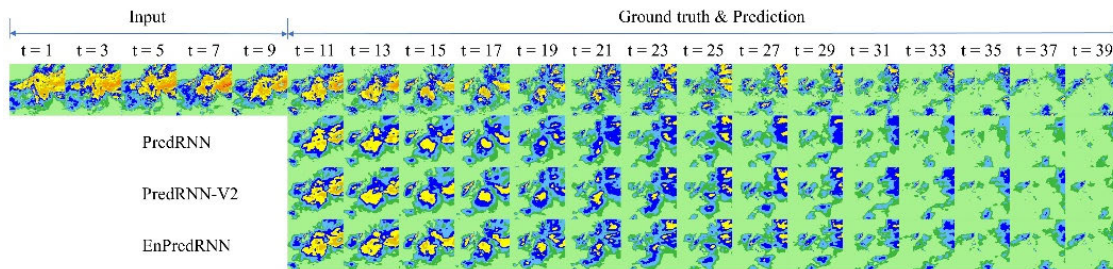


FIGURE 10. Qualitative results of the three models using the precipitation of the past hour to predict the precipitation of the next three hours. The radar echo dataset is more complex than the MovingMNIST and KTH. The prediction results on this dataset better visualize the gap in the predictive ability of different models.

into a mass of white as time goes by. These two models have poor long-term prediction ability. For PredRNN, PredRNN-V2 and EnPredRNN, all models can predict the shapes and positions of the numbers in the first 10 images (5 images are used here as the representative for the demonstration, the same as below) relatively accurately. When predicting for the middle 10 images, the clarity of the PredRNN prediction results decreases, and the clarity of the results of the other two models remains basically unchanged. The three models accurately predict the location of the digits in the images. When predicting for the last 10 images, PredRNN performs the worst, it predicts the appearance of the digits to be highly fuzzy, and the digit positions deviate from the ground truth. the shape of the digits predicted by PredRNN-V2 is slightly fuzzy, and the digit positions deviate from the ground truth. the prediction of the digit shapes and positions by EnPredRNN is the closest to the ground truth.

Since the long-term prediction ability of ConvLSTM and M-ConvGRU are much worse than that of PredRNN, PredRNN-V2, and EnPredRNN, we just choose the three models, PredRNN, PredRNN-V2, and EnPredRNN, for comparison in the experiments later. Table 4 shows the performance comparison between EnPredRNN and the other two models on the KTH dataset. From the table, it can be seen that the experimental results of PredRNN-V2 are better than PredRNN on MSE and PSNR, and worse than PredRNN on LPIPS and SSIM. EnPredRNN outperforms PredRNN and PredRNN-V2 in all the metrics, and it is the model with the best overall performance. Compared to the

MovingMNIST dataset, KTH contains more complex spatio-temporal patterns. The experimental results demonstrate the powerful and stable spatio-temporal modeling capability of the EnPredRNN.

The representative generated images of the three models are compared in Fig. 9. The image sequences used as samples contain relatively complex patterns of spatio-temporal variation. For comparison, the pictures are presented in the same way as in the MovingMNIST experiment. As can be seen from the figure, when predicting the first 10 frames of the image, the results of EnPredRNN are closest to the truth. Although PredRNN and PredRNN-V2 accurately get the trajectory of the human body, they lose part of the leg movement information. As the prediction proceeds, PredRNN and PredRNN-V2 can only get the rough outline of the human body, losing more detail information, and EnPredRNN gets the clearest image. It shows that EnPredRNN has excellent ability to model long-term dynamics.

2) EXPERIMENTS ON RADAR ECHO

Predicting the shape and motion trends of radar echoes in the future is an important application scenario for spatio-temporal predictive modeling. Compared to MovingMNIST and KTH, the Radar Echo dataset contains more complex spatio-temporal patterns. Accurate prediction of radar echoes in the future is a challenging task. In the paper, EnPredRNN, PredRNN and PredRNN-V2 are compared in the experiment on the real Radar Echo dataset. Thirty radar echo maps in the future are predicted using ten radar echo

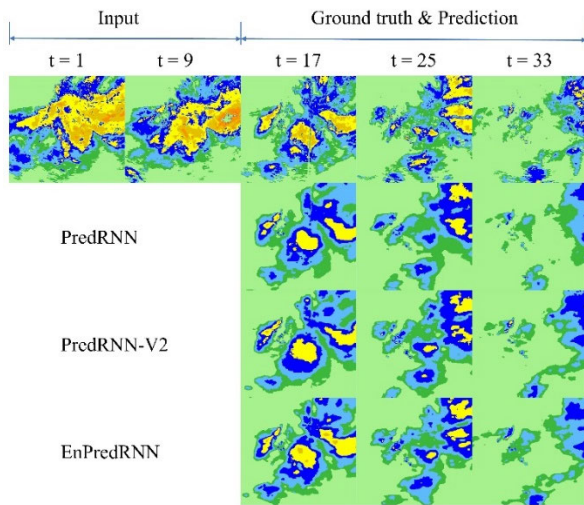


FIGURE 11. Presentation of the radar echo images in Fig. 10 at 3-frame intervals. Due to the complexity of the radar echo dataset, this figure is presented at larger intervals in order to show clearer details. The differences between the predictions of different models can be more clearly observed in the figure.

TABLE 5. CSI of each model at different thresholds and average SSIM over the dataset.

Model	CSI-10 $\uparrow$	CSI-20 $\uparrow$	CSI-30 $\uparrow$	SSIM $\uparrow$
PredRNN [7]	0.5464	0.3845	0.0819	0.8695
PredRNN-V2 [33]	0.5752	0.4285	0.0905	0.8705
EnPredRNN	0.5866	0.4523	0.1445	0.8751

TABLE 6. HSS for each model at different thresholds and MSE over the dataset.

Model	HSS-10 $\uparrow$	HSS-20 $\uparrow$	HSS-30 $\uparrow$	MSE $\downarrow$
PredRNN [7]	0.6643	0.5252	0.1425	166.31
PredRNN-V2 [33]	0.6904	0.5745	0.1570	158.43
EnPredRNN	0.6989	0.5961	0.2402	157.57

maps from the past. Tables 5 and 6 show the CSI and HSS for each model at three different thresholds of 10 dBZ, 20 dBZ and 30 dBZ and the average SSIM and MSE over the entire dataset. Fig. 10 shows the qualitative results of the three models. As in the previous approach, all sequence pictures are shown at 1-frame intervals. Fig. 11 presents the sequence pictures in Fig. 10 at larger intervals, showing clearer details.

As can be seen from Tables 5 and 6, EnPredRNN outperforms the other two models in terms of CSI and HSS at different thresholds, and also achieves the best average SSIM and MSE over the entire dataset. Specifically, when forecasting radar echoes with intensities above 10 dBZ, EnPredRNN shows minor enhancements over PredRNN and PredRNN-V2 on both CSI and HSS. It indicates that all three models are better at predicting light and moderate rainfall events in the future. As the threshold increases, the improvement of EnPredRNN over the other two models in terms of CSI and HSS becomes more pronounced. It indicates that EnPredRNN can better capture heavy precipitation events. The experimental

results of each model on SSIM and MSE show that the radar echo maps predicted by EnPredRNN are the most similar to the ground truth.

As can be seen in Fig. 10, all three models can basically predict the overall shape of the radar echoes and the specific distribution of the echoes of different intensities when predicting the first 10 images. When predicting the middle 10 images, the overall contours of the radar echoes predicted by PredRNN-V2 and PredRNN are somewhat different from the ground truth, and the shape of the echoes predicted by EnPredRNN is closest to the reality. In addition, some information of the strong echo region is lost in the prediction results of PredRNN-V2 and PredRNN, and the richest information of radar echoes of various intensities is retained in the results of EnPredRNN. The above experimental results show that when the prediction duration exceeds the certain threshold, PredRNN-V2 and PredRNN gradually show the lack of performance, while EnPredRNN can work better. When predicting the last 10 images, the radar echo shapes predicted by PredRNN-V2 and PredRNN are significantly different from the truth. The difference between the results predicted by EnPredRNN and the reality is smaller. Almost all the information of the strong echo region is lost in the images obtained by PredRNN-V2 and PredRNN. Some of the information of the strong echo region is retained in the results predicted by EnPredRNN. The prediction results of the last 10 images show that EnPredRNN has the strong ability to handle long sequences than PredRNN-V2 and PredRNN, and it can be used to prolong the spatio-temporal prediction. Fig. 11 shows the radar echo images in Fig. 10 every 3 frames. As can be seen from the figure, radar echoes with small intensity differences are not well distinguished in the prediction results of PredRNN-V2 and PredRNN, but are predicted to be of the same intensity. More details of the radar echo intensity differences are retained in the prediction results of EnPredRNN compared to the other models. In addition, from the last image, the prediction results of PredRNN-V2 and PredRNN are severely distorted and deformed, and the results of EnPredRNN are closest to the real radar echo distribution. It further demonstrates the effectiveness of EnPredRNN in long-term prediction tasks.

V. CONCLUSION AND DISCUSSIONS

The Enhanced PredRNN is designed on the basis of PredRNN in the paper, which can effectively extend spatio-temporal prediction period. Since the SA mechanism allows the model to better capture global dependencies in spatial information, the SA module is incorporated into the basic structural unit of PredRNN to form EnLSTM, which is the basic component of EnPredRNN. In addition, the ETM module and AGV structure are designed in the paper. The ETM module utilizes the correlation between spatio-temporal states to rationally aggregate the effective components of historical information as the input of the current time step, which extends the temporal sensory field of the model nodes and enables the model to better capture the temporal dependencies in

the sequence. The AGV structure provides the convenient pathway for the propagation of gradients, effectively copes with the gradient vanishing problem in recurrent neural networks, and enhances the ability of modeling long-term dependence of the model. In order to evaluate the performance, sufficient comparative experiments are conducted. The results of the ablation experiments show that our proposed series of improvements are beneficial to enhance the performance of the model. Experiments on MovingMNIST, KTH, and radar echo datasets validate the model's ability to handle long sequential information, demonstrating the model's ability to extend spatio-temporal predictions up to thirty time steps into the future while maintaining accuracy.

Although EnPredRNN with strong long-term prediction capability is proposed in the paper, there are still some points can be improved. To test the performance of EnPredRNN, comparative experiments were conducted only on the limited datasets. These results cannot ensure that EnPredRNN achieves better prediction results than other models in all cases. Therefore, we will conduct rigorous statistical tests in future work to draw more detailed and reliable conclusions. We will also conduct more interpretable studies towards the performance of EnPredRNN in the future, to deeply understand the internal operation mechanism of the model, to discover possible problems and shortcomings, and to point out the direction for the next step of improvement and optimization.

REFERENCES

- [1] L. Zhao, Y. Song, C. Zhang, Y. Liu, P. Wang, T. Lin, M. Deng, and H. Li, "T-GCN: A temporal graph convolutional network for traffic prediction," *IEEE Trans. Intell. Transp. Syst.*, vol. 21, no. 9, pp. 3848–3858, Sep. 2020.
- [2] J. Jing, Q. Li, X. Peng, Q. Ma, and S. Tang, "HPRNN: A hierarchical sequence prediction model for long-term weather radar echo extrapolation," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, May 2020, pp. 4142–4146.
- [3] Z. Chang, X. Zhang, S. Wang, S. Ma, and W. Gao, "STRPM: A spatiotemporal residual predictive model for high-resolution video prediction," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, New Orleans, LA, USA, Jun. 2022, pp. 13926–13935.
- [4] W. Zaremba, I. Sutskever, and O. Vinyals, "Recurrent neural network regularization," 2014, *arXiv:1409.2329*.
- [5] N. Srivastava, E. Mansimov, and R. Salakhudinov, "Unsupervised learning of video representations using LSTMs," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Jul. 2015, pp. 843–852.
- [6] X. Shi, Z. Chen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, "Convolutional LSTM network: A machine learning approach for precipitation nowcasting," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Jun. 2015, pp. 1–11.
- [7] Y. Wang, M. Long, J. Wang, Z. Gao, and P. S. Yu, "PredRNN: Recurrent neural networks for predictive learning using spatiotemporal LSTMs," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Long Beach, CA, USA, Dec. 2017, pp. 880–889.
- [8] G. Tan and Z. Wang, "Stability analysis of recurrent neural networks with time-varying delay based on a flexible negative-determination quadratic function method," *IEEE Trans. Neural Netw. Learn. Syst.*, early access, Nov. 3, 2023, doi: [10.1109/TNNLS.2023.3327318](https://doi.org/10.1109/TNNLS.2023.3327318).
- [9] J. Hu, G. Tan, and L. Liu, "A new result on H_∞ state estimation for delayed neural networks based on an extended reciprocally convex inequality," *IEEE Trans. Circuits Syst. II, Exp. Briefs*, vol. 71, no. 3, pp. 1181–1185, Mar. 2024.
- [10] Z. Liu, Y. Lin, Y. Cao, H. Hu, Y. Wei, Z. Zhang, S. Lin, and B. Guo, "Swin Transformer: Hierarchical vision transformer using shifted windows," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2021, pp. 9992–10002.
- [11] H. Zhou, S. Zhang, J. Peng, S. Zhang, J. Li, H. Xiong, and W. Zhang, "Informr: Beyond efficient transformer for long sequence time-series forecasting," in *Proc. AAAI Conf. Artif. Intell. (AAAI)*, Feb. 2021, pp. 11106–11115.
- [12] C. Wu, F. Wu, T. Qi, Y. Huang, and X. Xie, "Fastformer: Additive attention can be all you need," 2021, *arXiv:2108.09084*.
- [13] H. Hu, Z. Zhang, Z. Xie, and S. Lin, "Local relation networks for image recognition," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 3463–3472.
- [14] Z. Zhang, X. Lu, G. Cao, Y. Yang, L. Jiao, and F. Liu, "ViT-YOLO: Transformer-based YOLO for object detection," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2021, pp. 2799–2808.
- [15] W. Nie, N. Narodytska, and A. Patel, "RelGAN: Relational generative adversarial networks for text generation," in *Proc. Int. Conf. Learn. Represent.*, May 2019, pp. 1–20.
- [16] S. P. Singh, A. Kumar, H. Darbari, L. Singh, A. Rastogi, and S. Jain, "Machine translation using deep learning: An overview," in *Proc. Int. Conf. Comput., Commun. Electron. (Comptelix)*, Jul. 2017, pp. 162–167.
- [17] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, L. Kaiser, and I. Polosukhin, "Attention is all you need," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Long Beach, CA, USA, Dec. 2017, pp. 5999–6009.
- [18] R. Navares and J. L. Aznarte, "Predicting air quality with deep learning LSTM: Towards comprehensive models," *Ecological Informat.*, vol. 55, Jan. 2020, Art. no. 101019.
- [19] S. Hochreiter and J. Schmidhuber, "Long short-term memory," *Neural Comput.*, vol. 9, no. 8, pp. 1735–1780, Nov. 1997.
- [20] B. S. Zapata-Impata, P. Gil, and F. Torres, "Learning spatio temporal tactile features with a ConvLSTM for the direction of slip detection," *Sensors*, vol. 19, no. 3, p. 523, Jan. 2019.
- [21] E. Çiçek and S. Gören, "Smartphone power management based on ConvLSTM model," *Neural Comput. Appl.*, vol. 33, no. 13, pp. 8017–8029, Jul. 2021.
- [22] J. Su, W. Byeon, J. Kossai, F. Huang, J. Kautz, and A. Anandkumar, "Convolutional tensor-train LSTM for spatio-temporal learning," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Dec. 2020, pp. 13714–13726.
- [23] J. Chung, C. Gulcehre, K. Cho, and Y. Bengio, "Empirical evaluation of gated recurrent neural networks on sequence modeling," 2014, *arXiv:1412.3555*.
- [24] N. Ballas, L. Yao, C. Pal, and A. Courville, "Delving deeper into convolutional networks for learning video representations," 2015, *arXiv:1511.06432*.
- [25] L. Zhang, H. Yan, Y. He, S. Yao, S. Cao, and Q. Sun, "Spatiotemporal prediction of Alpine vegetation dynamic change based on a ConvGRU neural network model: A case study of the upper Heihe river basin in Northwest China," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 6957–6971, 2022.
- [26] Y. Zhang, L. Zhang, Y. He, S. Yao, W. Yang, S. Cao, and Q. Sun, "Analysis of the future trends of typical mountain glacier movements along the Sichuan-Tibet railway based on ConvGRU network," *Int. J. Digit. Earth*, vol. 16, no. 1, pp. 762–780, Mar. 2023.
- [27] W. He, T. Xiong, H. Wang, J. He, X. Ren, Y. Yan, and L. Tan, "Radar echo spatiotemporal sequence prediction using an improved ConvGRU deep learning model," *Atmosphere*, vol. 13, no. 1, p. 88, Jan. 2022.
- [28] X. Shi, Z. Gao, L. Lausen, H. Wang, D.-Y. Yeung, W.-K. Wong, and W.-C. Woo, "Deep learning for precipitation nowcasting: A benchmark and a new model," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Dec. 2017, pp. 5618–5628.
- [29] Q. Zeng, H. Li, T. Zhang, J. He, F. Zhang, H. Wang, Z. Qing, Q. Yu, and B. Shen, "Prediction of radar echo space-time sequence based on improving TrajGRU deep-learning model," *Remote Sens.*, vol. 14, no. 19, p. 5042, Oct. 2022.
- [30] W. Zhang, Y. Jiang, X. Song, B. Guoan, and R. Pang, "Application of XGBoost and TrajGRU to improve the accuracy of ECMWF wind forecasts," in *Proc. 14th Int. Conf. Graph. Image Process. (ICGIP)*, Jun. 2023, pp. 1101–1107.
- [31] Z. Zang, X. Bao, Y. Li, Y. Qu, D. Niu, N. Liu, and X. Chen, "A modified RNN-based deep learning method for prediction of atmospheric visibility," *Remote Sens.*, vol. 15, no. 3, p. 553, Jan. 2023.
- [32] J. Feng, J. Li, W. Zhong, J. Wu, Z. Li, L. Kong, and L. Guo, "Daily-scale prediction of Arctic sea ice concentration based on recurrent neural network models," *J. Mar. Sci. Eng.*, vol. 11, no. 12, p. 2319, Dec. 2023.

- [33] Y. Wang, H. Wu, J. Zhang, Z. Gao, J. Wang, P. S. Yu, and M. Long, "PredRNN: A recurrent neural network for spatiotemporal predictive learning," 2021, *arXiv:2103.09504*.
- [34] Y. Zhang and J. Wu, "Video extrapolation in space and time," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Tel Aviv, Israel, Oct. 2022, pp. 313–333.
- [35] Y. Zhou, C. Fu, Y. Zhai, J. Li, Z. Jin, and Y. Xu, "Identification of rice leaf disease using improved ShuffleNet V2," *Comput., Mater. Continua*, vol. 75, no. 2, pp. 4501–4517, 2023.
- [36] N. Ma, X. Zhang, H.-T. Zheng, and J. Sun, "ShuffleNet V2: Practical guidelines for efficient CNN architecture design," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Munich, Germany, Sep. 2018, pp. 122–138.
- [37] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, "MobileNetV2: Inverted residuals and linear bottlenecks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Salt Lake City, UT, USA, Jun. 2018, pp. 4510–4520.
- [38] A. Krizhevsky, I. Sutskever, and G. E. Hinton, "ImageNet classification with deep convolutional neural networks," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, Dec. 2012, pp. 1097–1105.
- [39] L. Huang, Z. Wang, and X. Fu, "Pedestrian detection using RetinaNet with multi-branch structure and double pooling attention mechanism," *Multimedia Tools Appl.*, vol. 83, no. 2, pp. 6051–6075, Jun. 2023.
- [40] H. Tang, H. Liu, D. Xu, P. H. S. Torr, and N. Sebe, "AttentionGAN: Unpaired image-to-image translation using attention-guided generative adversarial networks," *IEEE Trans. Neural Netw. Learn. Syst.*, vol. 34, no. 4, pp. 1972–1987, Apr. 2023.
- [41] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.
- [42] Z. Yi, H. Zhang, P. Tan, and M. Gong, "DualGAN: Unsupervised dual learning for image-to-image translation," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2868–2876.
- [43] T. Kim, M. Cha, H. Kim, J. K. Lee, and J. Kim, "Learning to discover cross-domain relations with generative adversarial networks," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Aug. 2017, pp. 1857–1865.



DALI WU was born in Hubei, China, in 1994. He received the master's degree from Qinghai University, in 2023. During the study period, he has participated in a number of national and provincial-level projects and received academic scholarships. He is currently a Teacher with the School of Computer and Information Science, Qinghai Institute of Technology. His research interests include spatial-temporal prediction and image processing.



JIAYANG KONG received the Ph.D. degree in statistics from CCNU, in 2023. He is currently pursuing the Ph.D. degree in public health with MUST.

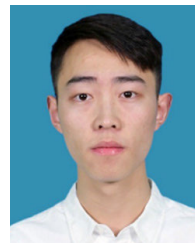
Upon graduating from CCNU, he joined the College of Computer Technology and Application, Qinghai University, where he is currently an Associate Professor, in 2024. He is currently an econometrician and an empirical microeconomist with broad disciplinary interests. His research interests include environmetrics, experimental psychology, behavioral economics, prediction, and data visualization.



ZHICHENG LI was born in Qinghai, China, in 1995. He received the master's degree from Northeast Forestry University, in 2020. During the study period, he has participated in a number of provincial-level projects and received some patents. He is currently a Teacher with the School of Computer and Information Science, Qinghai Institute of Technology. His research interests include SLAM, image processing, and electrical control.



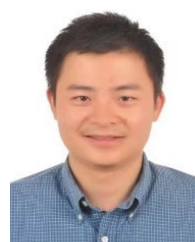
GUOJING ZHANG was born in Qinghai, China. She received the master's degree from Qinghai University, in 2017, where she is currently pursuing the Ph.D. degree. She participated in several national and provincial projects. She is now presiding over a School-Level Youth Fund and an Independent Project. She is currently a Teacher with the College of Computer Technology and Application, Qinghai University. Her current research interests include weather forecasting and grassland informatization.



HUAICONG ZHANG received the M.S. degree in cyberspace security from Southeast University, Nanjing, China, in 2023. He is currently working as a Teaching Assistant with the School of Computer and Information Science, Qinghai Institute of Technology. His current research interests include information security and artificial intelligence.



JING LIANG was born in Gansu, China. She received the Ph.D. degree, in 2023. During her study, she participated in several national and provincial projects and received academic scholarships. She is currently a Faculty Member with the School of Computer and Information Science, Qinghai Institute of Technology, China. Her research interests include complex networks and graph neural networks.



XING ZHANG was born in Hubei, China. He received the Ph.D. degree in computer science from Northwest A&F University, in 2017. He is currently an Associate Professor with the School of Computer and Information Science, Qinghai Institute of Technology. His research interests include deeply rooted in the exploration of intelligent information processing, data mining, and artificial intelligence.

...